Module 1 Assignment Report

Question 1: Technical Debt Diagnosis

Part 1: Matching each case to a technical debt type

Case A: The rounding logic of delivery time was changed. This change also hurt the favorite restaurants feature. This happened because of Entanglement. In the lecture this is called CACE. CACE means: Changing Anything Changes Everything. In a model, features are joined together inside. So a small change in one place can change other parts too, even if they look unrelated.

Case B: A marketing team was reading the model output table. The ML team did not know about this. This is called Undeclared Consumers. This means another team is using the model output without telling the ML team. This creates a hidden link. Because of this hidden link, the ML team cannot change the output safely.

Case C: The training pipeline has 14 shell scripts. There is no order or tool to manage them. This is called Configuration and Glue Code Debt. This happens when settings and steps are spread across many scripts. There is no single place to see all settings. Over time this becomes messy. This messy pipeline is called a pipeline jungle.

Part 2: One mitigation

We pick Case C for the mitigation. The fix is to use MLflow to track the pipeline. Instead of running 14 separate shell scripts with no record, each step can log its parameters and results using MLflow. Every run can also log a tag for the git commit. This gives one clear place to see all settings and results. This removes the mess of scattered scripts and makes the pipeline easy to understand and repeat.